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Evidence from the European Football Market

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Roadmap (Talk-First Design)

1. Motivation and Background
2. Mirrleesian Model of Optimal Taxation
3. Understanding the Transfer Market and Policies
4. Data
5. Theoretical Framework
6. Motivating Graphs
7. Regression Analysis
8. Findings and Policies
9. How it Motivates My Work

Roadmap

Question & Motivation

- **Question:** How do favorable tax policies induce migration for high-skilled individuals?
- Using data which tracks footballers' movement across leagues and countries, how did tax policies in Spain and Denmark affect migration decisions?
- Find notable shifts in demographic makeups of national leagues following policy changes, indicating that favorable policy leads to an influx of high skilled labor.

Why Football?

- Football contracts and by extension, movements, are easily tracable given it is a profession which is for entertainment
- Footballers at an elite level are certainly in the top end of the income distribution, thus leading them to be directly impacted by a tax cut for high skilled labor
- Football specific policy (Bosman Rule) provides another source of mobility variation, allowing for analysis of tax policy at multiple phases in the "football life cycle"

Contribution

- **What's new**: Implements a novel dataset of high-earning, high-skilled laborers' mobility patterns
- **Why it matters**: Allows for analysis of cross-country migration in the face of tax policy changes

Roadmap

Overview

- 1981 Paper: "Migration and Optimal Income Taxes"
- How to optimally tax citizens earning income abroad as to reduce waste
- How do taxes impact the distribution of agents?
- How do taxes play as incentives for agents?

Model Considerations

The three criteria concerning migration research, according to Mirrlees

1. Restriction to the group who does not migrate
   - Suboptimal as this does not account for those who wished to move but were constrained
   - Misses those who leave and still contribute to their country, a group which countries do not appear to be too concerned with
   - Restriction of taxes and corresponding benefits to voters, a confusing group as there are various definitions of voters
2. Definition of "Nationals"
   - Overly restrictive and presumptuous (i.e. exclusion of immigrants)
3. All humans
   - Not realistic as this definition would neglect welfare of the citizens of other countries

Foreign-Income Tax Model

- Consumer utility: $u(x, y, y', z, n)$
- Tax policy of the domestic government: $x = c(y, z) + y'$
- Foreign Income is taxed at a higher marginal tax rate than domestic when:

$$\frac{\partial}{\partial n} \frac{u_y(x, y, z, n)}{u_z(x, y, z, n)} < 0$$

  - note: case when untaxed foreign earnings y' are not present

Model, cont.

- When introducing y', agent chooses value to maximize:

$$u(c(y,z)+y',y,y',z,n)$$

- Envelope theorem yields:

$$\bar{u}_y = u_y(c+y',y,y',z,n)$$
$$\bar{u}_z = u_z(c+y',y,y',z,n)$$

$$s(c,y,y',z,n) = \frac{u_y}{u_z}$$

such that foreign income is taxed higher at the margin if:

$$s_n + s_{y'}\frac{\partial y'}{\partial n} > 0$$

- note: s is willingness to substitute domestic and foreign earnings
- n and y' play opposition to each other on s, with $s_n < 0$ and $s_{y'} > 0$
- Implies that a lower marginal tax rate on domestic income is reasonable with y'

Connecting to Kleven et. al

- Use this type of model to show how migration patterns vary by ability level of footballers
- Takes exogenous policy changes as opportunity to test intuition provided by model
- More concerned with migration responses to the tax policy than defining optimal policy

Roadmap

International Football, an Overview

- Football in Europe is a behemoth: 50+ leagues, 21,000 players
- Top Five Leagues: England, Spain, Germany, Italy, France
- €20.4bn in revenue from Top Five alone

Massive Player Transfer Market - €11bn in 2025

The Transfer Market

- Teams pay to purchase rights to players' services
- Two windows:
  1. Summer: June - September 1, seen as main window
  2. Winter: January
- System allows for player movement to be easily followed
- Contract numbers are commonly recorded, with approximations easily accessible

An Example of Transfers

- In 2025, Liverpool F.C. spent €520m during the summer breaking the Premier League record for most expensive transfer twice
- Importantly, not one of the seven incoming players are English or British
- In Liverpool's last match (at time of writing), they did not start a single British player, with only two senior British players being in the larger squad (20 players)
- How is this possible?

Roadmap

What is Their Goal?

- How does variation in top tax rates affect the migration of footballers in Europe?
- Exploit variation across multiple dimensions:
  1. Variation in top tax rates across countries and over time
  2. Preferential taxation schemes in specific countries
  3. Bosman Rule
- Utilize a Mirrleesian theoretical framework applied to a multinomial logit to determine player choices

Policy Overview - Bosman

- Bosman ruling occurred in 1995, overhauling much of the football status quo
- Prior to Bosman ruling:
  1. Three-player rule: no more than three foreign players per team could participate in any game in UEFA club competitions
  2. Transfer-fee rule: require a transfer fee even when a player became a free agent
- Bosman posed as a major hurdle to mobility both domestically and across countries
- Note: Foreign-player rule was still in some sort of effect, applying to non-Europeans after 1995

Policy Overview - Top Tax Schema

Spain:

- Beckham law: Flat tax of 24% for foreign workers relocating to Spain after Jan. 1, 2004
- Reduction from 45% at time of passing
- Can not have been a taxable resident of Spain in any of the preceding 10 years to be eligible

Denmark:

- Researchers' Tax Scheme: Flat tax of 25% for high income foreigners signing for employment in Denmark after June 1, 1991
- Reduction from 68% top tax rate at time of introduction
- Can not have been tax liable to Denmark for 3 years prior to scheme and must make at least €103,000

Roadmap

Baseline Model - Flexible Demand

- $N = \{n_1, n_2, \cdots, c_N\}$ countries possessing a continuum of domestic potential players
- Each player is endowed with footballing ability $a \geq 0$
- If they play, the agent generates value a for their club
- As in Mirrlees, before-tax wage is equal to ability, a

Baseline Model, cont.

- Player also assigned country of origin m and preferences $\mu_m = (\mu_{1m}, \mu_{2m}, \cdots, \mu_{Nm})$ for each location
- A player thus gets utility from country n by

$$u(a(1-\tau_{nm})) + \mu_{nm}$$

- where $u(\cdot)$ is increasing and τ_{nm} is the tax rate in n for a player from m
- IC condition is thus

$$u(a(1-\tau_{nm})) + \mu_{nm} \geq \max_{n'}\{u(a(1-\tau_{n'm})) + \mu_{n'm}$$

Baseline Model, cont.

- Define $g_m(\mu_m, a)$ as the joint distribution of players from m with ability a playing in n
- This measure of player supply is denoted as p_{nma} henceforth
- Total Measure of players in country n from country m with any ability level as

$$p_{nm}(1-\tau_{nm}) \equiv \int_0^\infty p_{nma}(a(1-\tau_{nm}))da$$

Baseline Model, cont.

- Countries may seperate tax rates such that $\tau_{nn} = \tau_{nd}$ and $\tau_{nm} = \tau_{nf}\ \forall\ n \neq m$
- Distributions by foreign and domestic can be defined as

Domestic:

$$p_{nd}(1-\tau_{nd}) \equiv \int_0^\infty p_{nda}(a(1-\tau_{nd}))da$$

Foreign:

$$p_{nf}(1-\tau_{nf}) \equiv \int_0^\infty p_{nfa}(a(1-\tau_{nf}))da$$

Baseline Model, cont.

Proposition 1:

Assuming that density $g_m(\mu_m, a)$ is smooth and positive everywhere on domain D, $p_{nda}, p_{nfa} > 0$ for all n, a and

1. p_{nda} is decreasing in τ_{nd} and unaffected by τ_{nf}
2. p_{nfa} is decreasing in τ_{nf} and unaffected by τ_{nd}

- States own-tax effect on number of both types of player is negative at all abilities
- Cross-tax effect is zero

Baseline Model, cont.

- Denote elasticities of players in a country n w.r.t. net-of-tax rate on players in country n

  Foreign:

$$\varepsilon_{nf} = \frac{dp_{nf}}{d(1-\tau_{nf})}\frac{1-\tau_{nf}}{p_{nf}}$$

  Domestic:

$$\varepsilon_{nd} = \frac{dp_{nd}}{d(1-\tau_{nd})}\frac{1-\tau_{nd}}{p_{nd}}$$

- Given approx. 85-90 percent of players are domestic, cutting rates on only foreign players can attract a large number of new players while having meager effects on preexisting foreign players

Adding Rigid Demand

Assumptions

- Assume each country's football market hires a continuum of measure one of players
- Players are hired by a continuum of clubs of measure one (each club hires one player)
- No new entry by clubs
- Population of potential native players in a country n has measure $P_n > 1$, meaning not all players can play in equilibrium
- Non-players work in a regular labor market with wage set to zero

Rigid Demand, cont.

Lemma 1:

In any equilibrium, within any country n, surplus $s_n \geq 0$ captured by each club is constant across all clubs and players in said country. Thus, before-tax wage paid out to a player of ability a in country n is $a - s_n$. No player with $a < s_n$ plays in country n.

- Form follows as before, except for before-tax income now being $a - s_n$

Rigid Demand, cont.

- Total number of players in a country n across ability levels is now represented as

  Domestic:

$$p_{nd}(s_n, 1-\tau_{nd}) \equiv \int_{s_n}^{\infty} p_{nda}((a-s_n)(1-\tau_{nd}))$$

  Foreign:

$$p_{nf}(s_n, 1-\tau_{nf}) \equiv \int_{s_n}^{\infty} p_{nfa}((a-s_n)(1-\tau_{nf}))$$

- Both functions decrease in s_n, but p_{ni} is increasing in $(1-\tau_{ni})$ for $i=(f,d)$

Rigid Demand, cont.

- Only general equilibrium differs, requiring

$$p_{nd}(s_n, 1-\tau_{nd}) + p_{nf}(s_n, 1-\tau_{nf}) = 1$$
$$\Rightarrow s_n = s_n(1-\tau_{nd}, 1-\tau_{nf})$$

- By inserting equilirbium surplus into $p_{nd}, p_{nda}, p_{nf}, p_{nfa}$, obtain relationships which are functions of tax rates

Rigid Demand, cont.

Proposition 2:

Assume that countries are small and that density $g_m(\mu_m, a)$ is smooth and positive everywhere on its domain D. Then, $p_{nda}, p_{nfa} > 0$ for all $a > s_n$, and:

1. $s_n(1-\tau_{nd}, 1-\tau_{nf})$ decreases with τ_{nd} and τ_{nf}
2. $p_{nda}(1-\tau_{nd}, 1-\tau_{nf})$ decreases with τ_{nd} at high abilities, increases with τ_{nd} at low abilities, and increases with τ_{nf} at all abilities
3. $p_{nfa}(1-\tau_{nd}, 1-\tau_{nf})$ decreases with τ_{nf} at high abilities, increases with τ_{nf} at low abilities, and increases with τ_{nd} at all abilities
4. $p_{nd}(1-\tau_{nd}, 1-\tau_{nf})$ decreases with τ_{nd} and increases with τ_{nf}
5. $p_{nf}(1-\tau_{nd}, 1-\tau_{nf})$ decreases with τ_{nf} and increases with τ_{nd}

Discussion of Model

- Sorting effects related to own-tax and cross-tax effects:
  1. Taxing foreign players no reduces the number of foreign players at all ability levels
  2. Due to equilibrium sorting, there is now a cross-tax effect of taxing one group on the other
- High ability players have $a \gg s_n$ such that $a - s_n \simeq a$ (response to taxation is basically the same in both specifications), thus mobility elasticity for high-ability players is the upper bound relevant for other high-income occupations with flexible labor demand

Roadmap

Bosman Ruling Cross-Country Correlations